

William Xin

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OBJECTIVE

Aspiring GIS Developer with a strong foundation in civil engineering and geospatial technologies. Skilled in database development, web application creation, and spatial analysis. Seeking to leverage my technical expertise and problem-solving skills to deliver innovative GIS solutions.

EDUCATION

Master of GIS– Penn State, Online GPA 3.9/4.0	May 2024 – August 2025
GIS Certificate – Penn State, Online GPA 4.0/4.0	May 2023 – May 2024
B. S Civil Engineering – Purdue University, West Lafayette, IN GPA 3.1/4.0	August 2017- May 2022

RELEVANT COURSEWORK:

Civil Engineering Courses

- Structural Analysis
- Transportation Engineering
- Reinforced Concrete Design
- Hydraulics
- Civil Engineering Materials
- Structural Steel Design

GIS and Programming Courses

- GIS Database Development
- GIS Programming and Software Development
- Geospatial Web Application Development
- Cloud and GIS Development
- Geographical Information Analysis
- Statistical Methods

TECHNICAL SKILLS

Programming: MATLAB, Python, HTML, CSS, JavaScript, R, MySQL, Git
GIS Software: ArcGIS, QGIS, Leaflet

PROJECTS & EXPERIENCE

Academic Projects:

- **Automated Classification Tool:** Created a Python-based ArcGIS script tool to classify and label graduated symbology classes for a GIS Programming and Software Development class.
- **EV Charging Station Locator:** Created a full-stack GIS web app hosted on AWS EC2 that visualizes Illinois EV charging stations with real-time updates from a Python service script.
- **Historical Map Digitization:** Digitized a 1920 Charlottesville, Virginia city map from a Sanborn paper map using ArcGIS Pro.

Personal Projects:

- **Interactive Web Map Development:** Designed an interactive web map visualizing U.S. Census data using the Leaflet API
- **3D Population Density Visualization:** Created a 3D population density map using ArcGIS Pro to analyze urban development patterns.
- **Illinois Elevation and Watershed Mapping:** Developed an elevation and watershed map for Illinois using advanced geospatial techniques in ArcGIS Pro.